

AI StudyBuddy API

An AI-Powered Smart Study Assistant

Project Presentation

Artificial Intelligence & Data Science

1. Introduction

- AI StudyBuddy API is an intelligent API designed to support students in learning.
- It connects student applications with AI-based study assistance.
- It can answer questions, summarize content, generate quizzes, and create study plans.

2. Problem Statement

- Students spend time searching for study materials and preparing revision content.
- Traditional learning methods may not provide personalized support.
- Students need quick, simple, and interactive academic assistance.

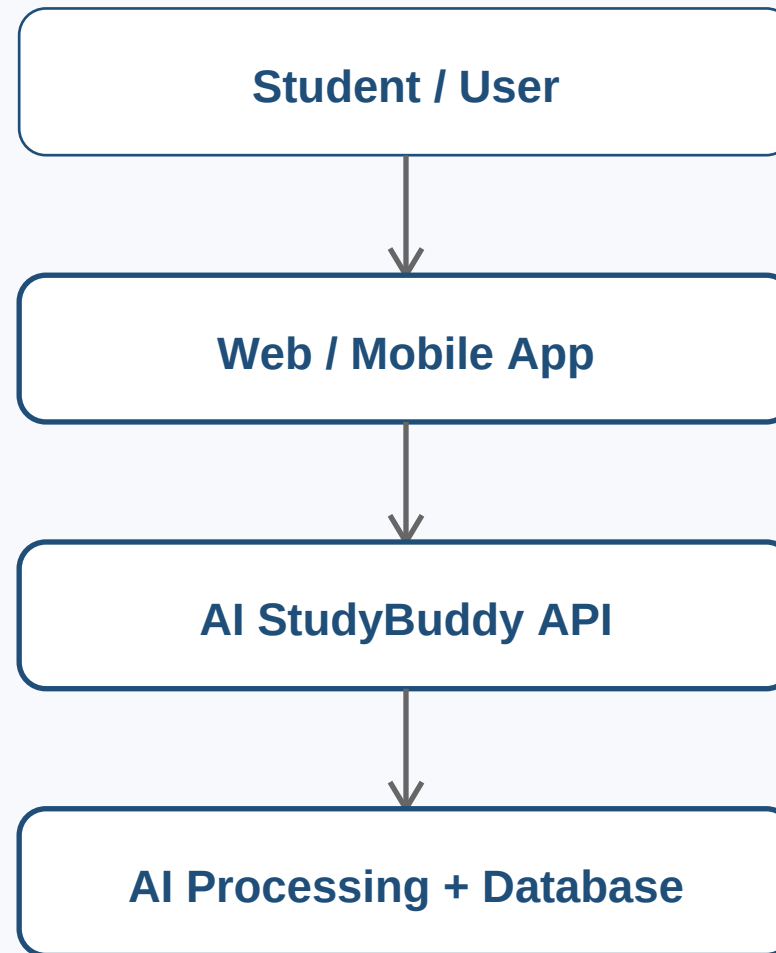
3. Objectives

- Provide instant academic assistance.
- Generate personalized study materials.
- Support quiz and practice-question generation.
- Help students organize their study schedule.
- Make AI learning features available through APIs.

4. Key Features

- AI Question & Answer
- Text Summarization
- Quiz and Question Generation
- Personalized Study Plan
- Simple Topic Explanation
- Learning Recommendations
- Progress Support

5. System Architecture



Personalized Response → Student

6. API Modules

- User Module – authentication and profile management
- Study Module – subjects, topics, and study plans
- Question Module – answers, quizzes, and practice questions
- Summary Module – concise study notes
- Recommendation Module – personalized suggestions
- Progress Module – learning and quiz progress

7. Example API Endpoints

- POST /api/v1/ask – Ask an academic question
- POST /api/v1/summarize – Summarize study content
- POST /api/v1/quiz – Generate quiz questions
- POST /api/v1/study-plan – Create a study plan
- GET /api/v1/progress – View learning progress

8. Technologies Used

- Frontend: HTML, CSS, JavaScript / React
- Backend: Python FastAPI or Node.js
- AI: Large Language Model (LLM)
- Database: MySQL / PostgreSQL / MongoDB
- API Format: REST API + JSON
- Deployment: Cloud or Local Server

9. Working Process

1. Student sends a request.
2. API authenticates and validates the request.
3. The request is processed by the AI layer.
4. AI generates a suitable learning response.
5. API returns the response to the application.
6. Learning information can be stored for future personalization.

10. Advantages & Future Scope

Advantages:

- Saves study time
- Provides quick academic support
- Supports personalized learning
- Helps revision and practice

Future Scope:

- Voice-based learning
- Multilingual support
- PDF/image-based question answering
- Adaptive quizzes and learning analytics

11. Conclusion

- AI StudyBuddy API provides a smart and flexible platform for student learning.
- It combines AI capabilities with REST API technology.
- The system can support question answering, summarization, quizzes, and study planning.
- It can be extended into web, mobile, and educational applications.